P-102 Detailed Equipment Datasheet

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P-102 is an AF-450C centrifugal process pump installed in 2018 for CDU-4 crude transfer.

Specification

Field	Value
Manufacturer	ApexFlow
Model	AF-450C
Capacity	560 m3/h
Design pressure	50 bar
Approved maximum operating pressure	40 bar
Normal operating pressure	35 bar
Normal service temperature	190 °C

Drive and instrumentation

The pump package includes a dedicated motor and independent process measurements.

Connected equipment

Tag	Role
M-102	450 kW, 6.6 kV motor
PT-102	Discharge pressure
TT-102	Discharge temperature
FT-210	Crude flow
PSV-102	Protection; set at 47.5 bar
XV-410	Crude-feed isolation

Operating envelope

The three pressure values have different engineering meanings.

Distinctions

- 50 bar = design pressure.
- 40 bar = approved internal operating maximum.
- 47.5 bar = PSV protection set point.
- 35 bar = normal operating basis.

Maintenance context

P-102 has recurring seal, vibration and corrosion observations.

Work orders

Date	WO	Activity
2022-08-14	WO-3310	Seal replacement
2024-03-12	WO-4102	Condition inspection
2024-11-04	WO-4531	Seal replacement
2025-04-16	WO-5012	Vibration inspection
2025-10-12	WO-5521	Corrosion inspection
2026-07-16	WO-6021	Integrity review

Current condition

The 2026 package requires a multi-source review.

Current evidence

- Pressure 42 bar: +2 bar above approved maximum.
- Corrosion 3.8 mm: above engineering-review trigger.
- Vibration 7.2 mm/s: above high-concern threshold.
- PT-102 verification remains open.
- Human engineering approval is required.